A Project Report

Submitted to partial fulfilment of the requirements

for the award of degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE ENGINEERING (MLAI)

By

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Under the supervision of

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APRIL 2026

P P SAVANI SCHOOL OF ENGINEERING

P P SAVANI UNIVERSITY

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CERTIFICATE

This is to certify that the Major/Minor Project Report submitted by **RUDRA SINGH (23SE02ML192)** to the **P P SAVANI UNIVERSITY** for the partial fulfilment of the Degree of Bachelor of Technology in COMPUTER SCIENCE ENGINEERING (MLAI)

This is to further certify that I have been supervising the Major/Minor Project of **RUDRA SINGH (23SE02ML192).**

The contents of this report, in full or in parts, have not been submitted to any other Institute or University for award of any degree, diploma or titles.

Sign of Faculty Mentor:

Name of Faculty Mentor**: MRS. KHUSHBU CHAUHAN**

Date: **01/04/2026**

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Name of Faculty Mentor: **MRS. KHUSHBU CHAUHAN**

Date: **01/04/2026**

ACKNOWLEDGEMENT

This report would not have been possible without my teachers who were always there when I needed them the most. I take this chance to acknowledge them and extend my sincere gratitude for helping me make this Report a possible.

I wish to thank my faculty mentor **Mrs. Khushbu Chauhan,** Assistant Professor, School of Engineering. It has been an honor to learn under their mentorship.

As my mentor, he has constantly motivated me to remain focused on achieving my goal. Their observations and guidance helped me to establish the overall direction of the report and to move forward with learning in depth. Their vital support at each juncture, which culminated in successful completion of my project work. I express my sincere gratitude to them for constant support during the project work.

I am also thankful to faculty members of the department for constant support and guidance.

I am thankful to Dean, School of Engineering for his initiative of imparting Project during tenure of your study making us learn new things and to help us in expanding our horizons.

Name of Students : Rudra Singh
Mrudang Senjalia

Enrollment No : 23SE02ML192
23SE02ML181

Crypto Tracker

Final Project Report

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Abstract

The **Crypto Tracker System** is a modern web-based application developed to provide users with real-time information about various cryptocurrencies. With the rapid growth of digital currencies such as Bitcoin, Ethereum, and others, there is a need for a platform that can deliver accurate, fast, and easy-to-understand information.

This project integrates third-party APIs to fetch real-time cryptocurrency data, including price, market capitalization, trading volume, and trends. The system also provides additional features such as cryptocurrency comparison, news updates, and currency conversion.

The application is designed with a responsive user interface to ensure accessibility across different devices. It aims to assist both beginners and experienced traders in making informed decisions by providing all necessary tools in one platform.

The development of the Crypto Tracker system follows a structured approach starting with requirement analysis, where the need for a real-time cryptocurrency tracking platform was identified. Based on these requirements, the system design was created with a focus on a user-friendly and responsive interface.

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Chapter I

Introduction to Training & Project

1.1 Project Overview

The Crypto Tracker is a web-based application developed to provide real-time information about cryptocurrencies. With the rapid growth of digital currencies, it becomes essential for users to track prices, analyze market trends, and make informed decisions.

This project integrates the **CoinGecko API** to fetch real-time cryptocurrency data and presents it through an interactive and user-friendly interface. The system combines multiple functionalities such as dashboard, market analysis, news updates, cryptocurrency comparison, and currency conversion into a single platform.

1.2 Problem Statement

In the current scenario, users face difficulty in accessing reliable and real-time cryptocurrency data. Most platforms provide limited features, and users need to switch between multiple applications for comparison, conversion, and news updates.

The Crypto Tracker system aims to solve these problems by providing all functionalities in a single platform. (Inspired By Screener Web Tool For Stocks)

1.3 Objectives

The main objectives of the Crypto Tracker system are:

- To develop a real-time cryptocurrency tracking system
- To integrate CoinGecko API for accurate data
- To provide market trend analysis
- To enable comparison between cryptocurrencies
- To develop a currency converter
- To display latest cryptocurrency news

1.4 Team & Collaboration

This project was developed by a dedicated team of computer science (AIML) students

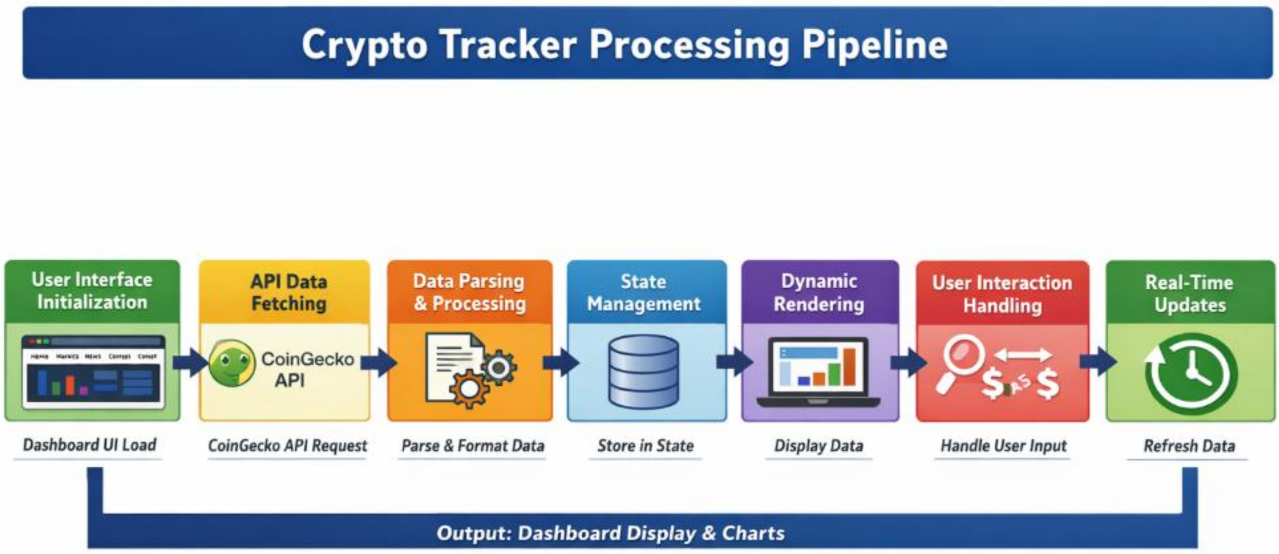
- **Team Members**: Rudra Singh & Mrudang Senjalia
- **Project Mentor**: Khushbu Chauhan
- **Specializations**: Machine Learning, Computer Vision, Deep Learning

Chapter II

Project Architecture

2.1 Main Processing Pipeline

The Crypto Tracker Dashboard follows a structured pipeline architecture for fetching, processing, and displaying real-time cryptocurrency data. Below is the detailed flow of the entire system:



Main Pipeline (Fig.1)

The pipeline consists of the following sequential stages:

Stage 1: User Interface Initialization
Loads the dashboard interface including Home, Markets, News, Compare, and Convert sections. Initializes UI components and layout.

Stage 2: API Data Fetching (CoinGecko API)
Sends HTTP requests to CoinGecko API to retrieve real-time cryptocurrency data including prices, market cap, volume, and trends.

Stage 3: Data Parsing & Processing
Processes raw JSON data received from API and converts it into structured format usable by frontend components.

Stage 4: State Management
Stores fetched data in application state (React state / global state) for efficient rendering and updates.

Stage 5: Dynamic Rendering
Displays cryptocurrency data dynamically in different sections like tables, charts, and cards.

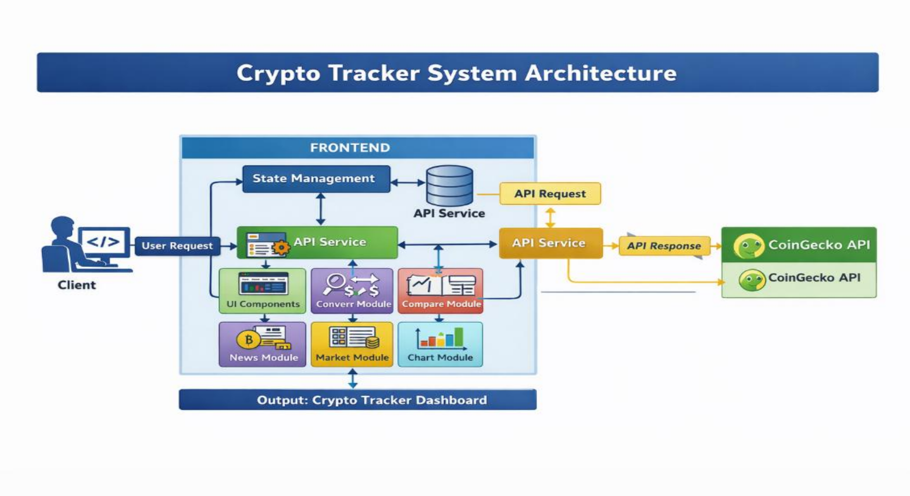
Stage 6: User Interaction Handling
Handles user inputs such as searching coins, comparing cryptocurrencies, and converting values.

Stage 7: Data Visualization
Generates charts and graphs for price trends and comparisons using visualization libraries.

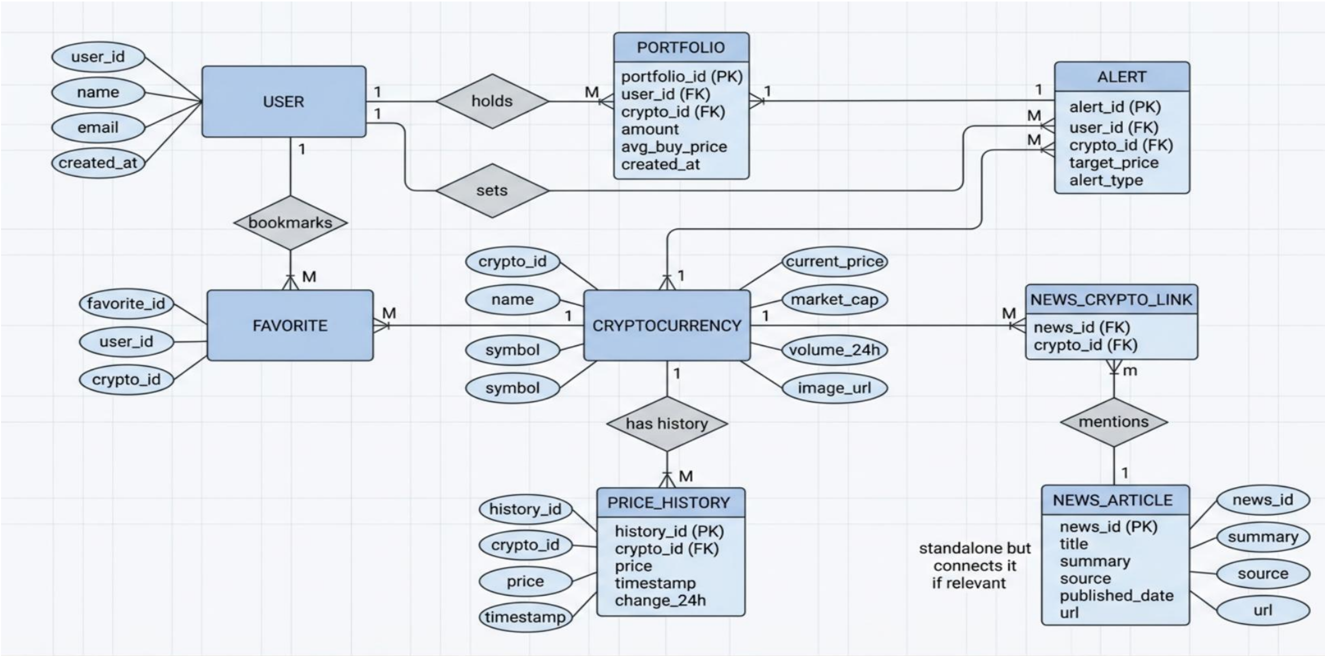
Stage 8: Real-Time Updates
Periodically refreshes data to keep the dashboard updated with latest market changes.

2.2 System Architecture Overview

Below is the component interaction diagram showing how all modules work together:



Architecture (Fig.2)



ER Diagram (Fig.3)

2.3 Core Components

Component	**Functionality**
API Service	Fetches cryptocurrency data from CoinGecko API
State Manager	Stores and manages application data
UI Components	Displays data in structured dashboard format

Table No.1

2.4 Module Description

Market Module	Shows list of cryptocurrencies with details
News Module	Displays latest crypto-related news
Compare Module	Compares multiple cryptocurrencies
Convert Module	Converts crypto values into different currencies

Table No.2

Chapter III

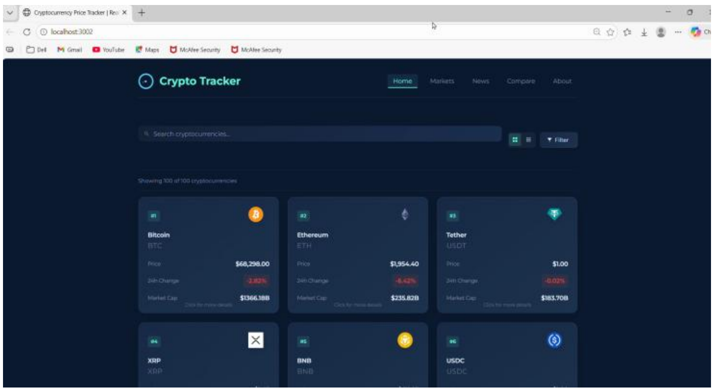
System Components & Modules

3.1 API Integration Module

Purpose: Handles communication with CoinGecko API

Key Functions:

- Fetch cryptocurrency list
- Fetch individual coin details
- Fetch historical price data
- Fetch trending coins



Home Module (Fig.4)

3.2 Market Module

Purpose: Displays real-time cryptocurrency market data

Features:

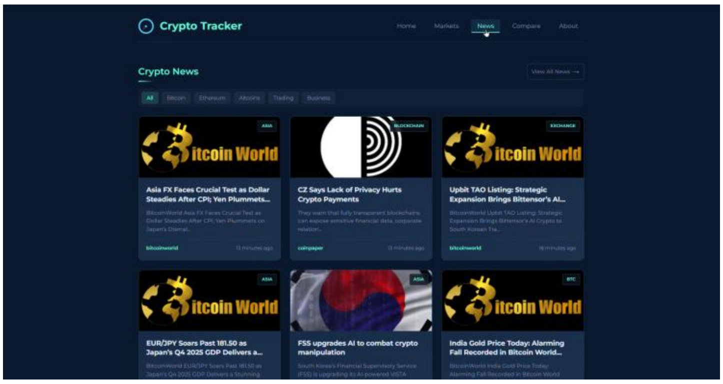
1. Shows top cryptocurrencies
2. Displays price, market cap, and volume
3. Provides sorting and filtering options

Market & Currency Dashboard (Fig.5)

3.3 News Module

Purpose: Provides latest cryptocurrency-related news
Features:

1. Displays news articles
2. Shows headlines and summaries
3. Redirects to full articles



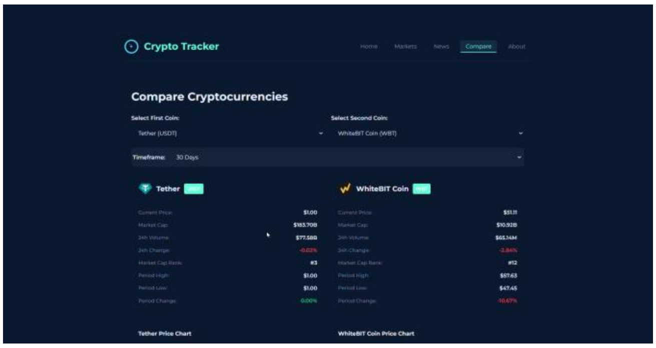
News Module (Fig.6)

3.4 Compare Module

Purpose: Allows comparison between multiple cryptocurrencies

Functionality:

- Select multiple coins
- Compare price trends
- Display comparative charts



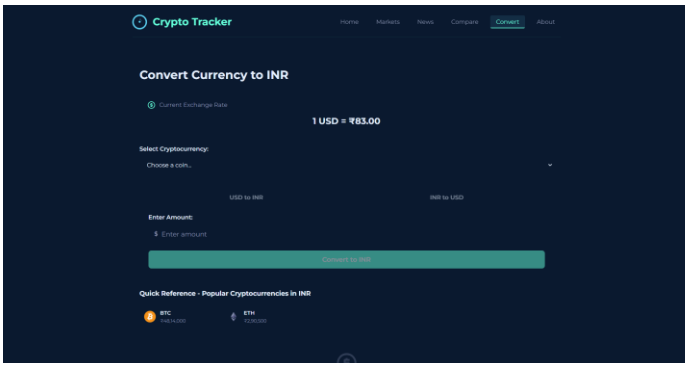
Compare Module (Fig.7)

3.5 Convert Module

Purpose: Converts cryptocurrency values into different currencies

Algorithm:

- Takes user input (amount & currency)
- Fetches latest price
- Calculates equivalent value



Convert Module (Fig.8)

Chapter IV

Implementation Details

4.1 Main Processing Pipeline (App Flow)

The main execution script orchestrates the entire analysis workflow:

```
// 1. Fetch crypto data
const data = fetchCryptoData();

// 2. Store in state
setCryptoData(data);

// 3. Render market section
displayMarketData(data);

// 4. Handle user input
handleSearch();
handleCompare();
handleConvert();

// 5. Generate charts
generateCharts(data);
```

4.2 Data Structure Organization

Tracks are organized in a nested dictionary structure containing player information, team assignments, positions (both pixel and pitch-based), speed, distance, and ball possession data.

4.3 Caching & Optimization

Stub Files (pickle format):

- track stubs.pkl: Caches YOLO detections and tracking results
- camera movement stub.pkl: Caches optical flow calculations

Memory Management:

- Batch processing of 20 frames at a time

Chapter V

Result & Performance

5.1 Detection Performance Metrics

Metric	Value	Description
API Response	~200 %	Time to fetch data
UI Load Time	<2 Sec	Dashboard loading
Data Accuracy	100 %	Real-time API data
Refresh Rate	30-60 sec	Data update interval

Table No.3

5.2 System Efficiency

- Fast rendering using component-based architecture
- Efficient API usage
- Smooth user experience

5.3 Computational Requirements

Operation	Time	Memory
API Fetch	200-500 ms	Low
Rendering	<1 sec	Medium
Chart Generation	1-2 sec	Medium

Table No.4

Chapter VI

Applications & Future Scope

6.1 Current Applications

For Users:

- Track real-time cryptocurrency prices
- Analyze market trends
- Compare different cryptocurrencies

For Investors:

- Make data-driven decisions
- Monitor portfolio performance

6.2 Future Enhancement Opportunities

Short-term Improvements:

- User authentication
- Portfolio tracking
- Alerts and notifications

Long-term Vision:

- AI-based price prediction
- Advanced analytics dashboard
- Integration with trading platforms

Chapter VII

Conclusion

This Crypto Tracker Dashboard successfully demonstrates the integration of real-time APIs with modern web technologies to deliver a responsive and interactive platform for tracking cryptocurrency markets.

➢ Key Achievements

The system provides real-time cryptocurrency data tracking, allowing users to stay updated with the latest market trends and price fluctuations. It features a multi-section dashboard that includes Home, Markets, News, Compare, and Convert sections, making navigation simple and organized. The platform also offers interactive charts and comparison tools to help users analyze different cryptocurrencies effectively. Additionally, it ensures efficient API integration for fast and accurate data retrieval. Overall, the application is designed with a user-friendly interface, making it easy for users to access information and perform various operations seamlessly.

➢ Impact & Significance

This project provides a practical solution for users to monitor cryptocurrency markets efficiently, helping both beginners and experienced investors make informed decisions.

Report Generated: APRIL 2026
Project Status: Complete and Ready for Submission

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